

Name _____ Date _____ Period _____

Domain and Range: All Three Notations

Algebra II | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write the domain and range of a function from a graph, a table, or a set of ordered pairs using a verbal description, inequality notation, set notation, and interval notation.

Vocabulary

Domain — the set of all _____ values, the _____-values.

Range — the set of all _____ values, the _____-values.

Set notation — uses _____ with the bar | , read "such that."

Interval notation — uses _____ if an endpoint is included, _____ if it is excluded.

Inequality notation — writes domain or range using _____ instead of interval or set notation.

Part 1 — From a Set of Ordered Pairs

Least to greatest, and each value written only once.

Example 1. $\{(-4, -6), (0, -1), (2, 4), (3, -7), (7, 7)\}$

Domain: { _____ }

Range: { _____ }

You Try 1. $\{(-5, 3), (-1, 0), (2, 3), (6, -8)\}$

Domain: { _____ }

Range: { _____ }

Part 2 — From a Table

Example 2.

x	-3	-1	0	2	5
f(x)	9	1	0	4	25

Domain: { _____ } Range: { _____ }

You Try 2.

x	-2	0	1	4
f(x)	7	7	-3	10

Domain: { _____ } Range: { _____ }

Part 3 — The Three Notations

The first row is done for you. Fill in the rest.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	$-\infty < x < \infty$	$\{x \mid x \in \mathbb{R}\}$	$(-\infty, \infty)$
all reals greater than or equal to 0	$x \geq 0$	_____	_____
all reals less than 5	$x < 5$	_____	_____

all reals from -3 to 3, including both	$-3 \leq x \leq 3$	_____	_____
just the numbers -2, 0, and 4	(none)	_____	_____

Key idea: bracket [] = endpoint _____. Parenthesis () = endpoint _____. ∞ always takes a _____.

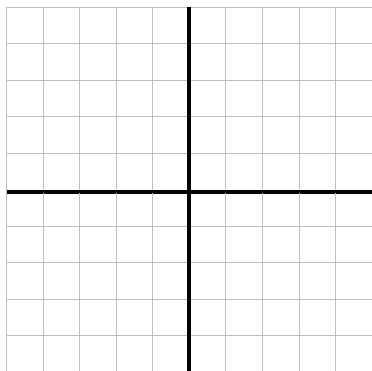
You Try 3. all reals greater than -8 $\rightarrow$ inequality _____ interval _____

You Try 4. $y \leq 12 \rightarrow$ in words _____ interval _____

Part 4 — From a Graph

Each square is 1 unit. The axes are the two heavy lines, and both grids run -5 to 5.

A. Discrete. Plot (-4, 2), (-2, -3), (1, 4), (3, -1).

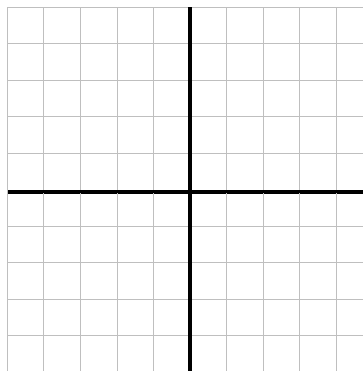


Domain (set): _____

Range (set): _____

Continuous or discrete? _____

B. Continuous. Graph $y = x + 1$ from $x = -4$ to $x = 3$.



Domain (inequality): _____

Domain (interval): _____

Range (interval): _____

Part 5 — Continuous or Discrete?

Write continuous or discrete.

- The number of students in each class period _____
- The outside temperature over one full day _____
- The total cost of n concert tickets at \$45 each _____

Exit Ticket

Given $\{(1, 5), (3, 5), (4, -2)\}$ write the domain and range in set notation, then say whether interval notation could be used.

Domain _____ Range _____

Could interval notation be used? _____